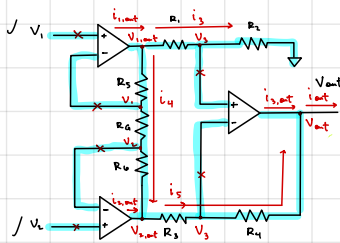


Instrumentation amplifier



input signals to circuit

* power supplies per op-amp

↳ LM741

offset null	1	8	NC (no connect, leave floating)
inverting in	2	7	V ⁺ (power supply)
non-inverting in	3	6	output
(GND) V ⁻	4	5	offset null

Calculate transfer function

$$V_{out} = F_T V_{in}$$

Voltage drops across each R

- (1) $V_{1,out} - V_3 = i_3 \cdot R_1$ ✓
- (2) $V_3 = i_3 \cdot R_2$ ✓
- (3) $V_{2,out} - V_3 = i_5 \cdot R_3$ ✓
- (4) $V_3 - V_{out} = i_5 \cdot R_4$ ✓
- (5) $V_{1,out} - V_1 = i_4 \cdot R_5$ ✓
- (6) $V_1 - V_2 = i_4 \cdot R_6$ ✓
- (7) $V_2 - V_{2,out} = i_4 \cdot R_6$ ✓

7 eqs & 6 unknowns: $V_{1,out}, V_3, i_3, V_{2,out}, i_5, i_4$

Solve for: V_{out} into V_1, V_2 & R 's

technique: (1) eliminate all dependence on current (2) remove dep. on $V_{1,out}, V_{2,out}$ & V_3

$$V_{1,out} = V_3 \left(1 + \frac{R_1}{R_2}\right) \quad (2a)$$

$$(6) \rightarrow (5) \quad V_{1,out} - V_1 = (V_1 - V_2) \frac{R_5}{R_6}$$

$$V_{1,out} = V_1 \left(1 + \frac{R_5}{R_6}\right) - V_2 \frac{R_5}{R_6} \quad (5a)$$

$$(6) \rightarrow (7) \quad V_2 - V_{2,out} = (V_1 - V_2) \frac{R_6}{R_6}$$

$$V_{2,out} = (V_1 - V_2) \frac{R_6}{R_6} + V_2$$

$$V_{2,out} = V_1 \left(1 + \frac{R_6}{R_6}\right) - V_2 \frac{R_6}{R_6} \quad (7a)$$

Solve (2a) for V_3

$$V_{1,out} = V_3 \left(1 + \frac{R_1}{R_2}\right) \quad (2a)$$

(2a) → (4a) eliminate V_3

$$V_{out} = V_{1,out} \cdot \frac{R_3}{R_1 + R_2} \left(1 + \frac{R_4}{R_3}\right) - \frac{R_4}{R_3} V_{2,out} \quad (4b)$$

(5a) → (4b) eliminate $V_{1,out}$

$$V_{out} = \left[V_1 \left(1 + \frac{R_5}{R_6}\right) - V_2 \frac{R_5}{R_6} \right] \frac{R_3}{R_1 + R_2} \left(1 + \frac{R_4}{R_3}\right) - \frac{R_4}{R_3} V_{2,out} \quad (4c)$$

(7a) → (4c) to eliminate $V_{2,out}$

$$V_{out} = \left[V_1 \left(1 + \frac{R_5}{R_6}\right) - V_2 \frac{R_5}{R_6} \right] \frac{R_3}{R_1 + R_2} \left(1 + \frac{R_4}{R_3}\right) - \frac{R_4}{R_3} \left[V_1 \left(1 + \frac{R_6}{R_6}\right) - V_2 \frac{R_6}{R_6} \right]$$

$$V_{out} = V_1 \left(1 + \frac{R_5}{R_6}\right) \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} + V_1 \cdot \frac{R_6}{R_6} \cdot \frac{R_4}{R_3}$$

$$- V_2 \frac{R_5}{R_6} \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} - V_2 \cdot \left(1 + \frac{R_6}{R_6}\right) \cdot \frac{R_4}{R_3}$$

$$V_{out} = V_1 \left(1 + \frac{R_5}{R_6}\right) \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} + V_1 \cdot \frac{R_6}{R_6} \cdot \frac{R_4}{R_3}$$

$$- V_2 \frac{R_5}{R_6} \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} - V_2 \cdot \left(1 + \frac{R_6}{R_6}\right) \cdot \frac{R_4}{R_3}$$

Check, if $R_1 = R_2, R_4 = R_5 \& R_3 = R_6$ do we get match from Justin Bauer notes?

$$\begin{aligned} V_{out} &= V_1 \left(1 + \frac{R_5}{R_6}\right) \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} + V_1 \cdot \frac{R_6}{R_6} \cdot \frac{R_4}{R_3} - V_2 \frac{R_5}{R_6} \left(1 + \frac{R_4}{R_3}\right) \frac{R_3}{R_1 + R_2} - V_2 \cdot \left(1 + \frac{R_6}{R_6}\right) \cdot \frac{R_4}{R_3} \\ &= V_1 \cdot \frac{R_3}{R_1 + R_2} \left(1 + \frac{R_5}{R_6}\right) \left(1 + \frac{R_4}{R_3}\right) + V_1 \cdot \frac{R_6}{R_6} \cdot \frac{R_4}{R_3} - V_2 \frac{R_3}{R_1 + R_2} \left(1 + \frac{R_5}{R_6}\right) \frac{R_3}{R_6} - V_2 \cdot \left(1 + \frac{R_6}{R_6}\right) \cdot \frac{R_4}{R_3} \\ &= V_1 \left[\frac{R_3}{R_1 + R_2} \left(1 + \frac{R_5}{R_6}\right) \left(1 + \frac{R_4}{R_3}\right) + \frac{R_6}{R_6} \cdot \frac{R_4}{R_3} \right] - V_2 \left[\frac{R_3}{R_1 + R_2} \left(1 + \frac{R_5}{R_6}\right) \frac{R_3}{R_6} + \left(1 + \frac{R_6}{R_6}\right) \frac{R_4}{R_3} \right] \\ &= V_1 \left[\frac{R_3}{R_1} \left(1 + \frac{R_5}{R_6}\right) + \frac{R_6}{R_6} \cdot \frac{R_4}{R_3} \right] - V_2 \left[\frac{R_3}{R_1} \frac{R_5}{R_6} + \frac{R_6}{R_6} \left(1 + \frac{R_6}{R_6}\right) \right] \\ &= V_1 \cdot \frac{R_3}{R_1} \left(1 + 2 \frac{R_5}{R_6}\right) - V_2 \frac{R_3}{R_1} \left(1 + 2 \frac{R_5}{R_6}\right) \end{aligned}$$

$$V_{out} = (V_1 - V_2) \frac{R_3}{R_1} \left(1 + 2 \frac{R_5}{R_6}\right) \quad \text{it matches!!}$$

Should be

$$V_{out} = (V_1 - V_2) \frac{R_3}{R_1} \left(1 + 2 \frac{R_5}{R_6}\right) \quad \checkmark$$

continued

$$V_{out} = (V_1 - V_2) \frac{R_3}{R_1} \left(1 + 2 \frac{R_5}{R_6}\right)$$

We can further make $R_1 = R_2 = R_3 = R_6 = 100 \Omega$ → chosen at feedback

$$V_{out} = (V_1 - V_2) \left(1 + \frac{40k\Omega}{100\Omega}\right) \quad R_6 \text{ was chosen to be } 100 \Omega$$

$$V_{out} = (V_1 - V_2) \left(1 + \frac{400}{1}\right)$$

$$V_{out} = (V_1 - V_2)(401)$$

So, approximately a gain of ~400